

Markscheme

Mathematics: analysis and approaches

Practice question bank

164. (a)

Consider the equation $z^4 = -16$.

Find the root with both a positive real part and a positive imaginary part, in the form $a + bi$, correct to 3 significant figures.

$$-16 = 16(\cos 180^\circ + i \sin 180^\circ) \quad (M1)$$

4th roots have modulus 2 and arguments $\frac{180^\circ + 360^\circ k}{4}$ for $k = 0, 1, 2, 3$ (M1) A1

the root with both parts positive has argument 45° (M1)

$$2(\cos 45^\circ + i \sin 45^\circ) \approx 1.41 + 1.41i \quad \text{A1}$$

[6 marks]